Lecture 5.2 – Filtering, Robustness, and Final Controller Testing

Week 5: Making the Motor Controller More Reliable

Learning Objectives

By the end of this lecture, you should be able to:

- explain why measured RPM may be noisy;
- apply a simple exponential moving-average filter;
- describe the tradeoff between smoothing and delay;
- define controller robustness in this project;
- use measured data to choose a final controller;
- prepare for final project testing and presentation.

1. Why Filter the RPM Measurement?

The encoder gives pulses, not a perfectly smooth speed measurement.

The calculated RPM can vary because of:

- small number of encoder counts per sample;
- quantization;
- motor vibration;
- timing variation;
- electrical noise;
- mechanical load changes.

If the controller reacts too strongly to noisy measurements, the PWM command may also become noisy.

> Filtering can make the measured RPM smoother, but it can also add delay.

2. Exponential Moving-Average Filter

A simple filter is

$$\boxed{y_f[k] = \alpha y[k] + (1-\alpha) y_f[k-1].}$$

where

- $y[k]$ is the current raw RPM;
- $y_f[k]$ is the filtered RPM;
- $0 < \alpha \leq 1$.

If

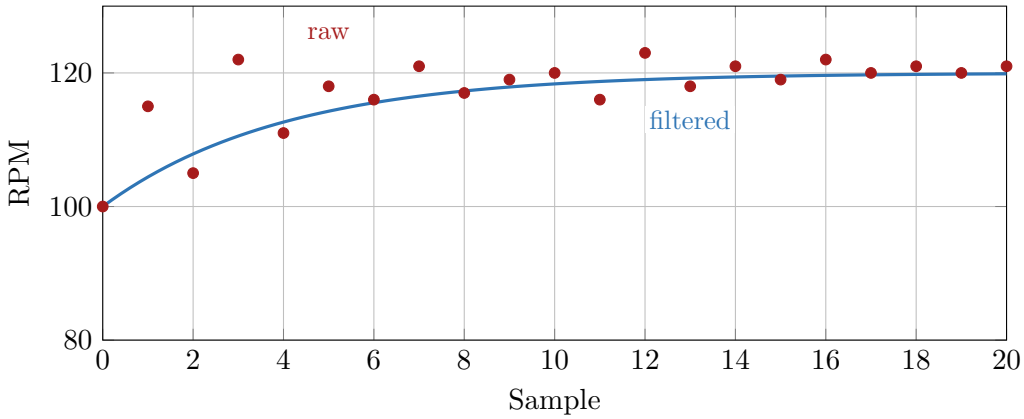
$$\alpha = 1,$$

there is no filtering.
If

$$\alpha$$

is smaller, the signal becomes smoother but slower to respond.

3. Filter Examples



4. Choosing α

Typical values:

α	Behavior
1.0	no filtering
0.7	light filtering
0.4	moderate filtering
0.2	strong filtering, more delay

For this class, start with

$$\alpha = 0.5$$

or

$$\alpha = 0.7.$$

Do not use filtering to hide a bad controller. Filtering should reduce measurement noise, not cover up unstable behavior.

5. Filtering Tradeoff

Filtering helps when the measured speed is noisy.

However, filtering also means the controller sees a delayed version of the real speed.

Too much filtering can cause:

- slow response;
- delayed disturbance correction;
- overshoot;
- poor tracking during speed changes.

A good filter is smooth enough to reduce noise but fast enough to preserve useful feedback.

6. Robustness in This Course

In this course, a robust motor controller should:

- reach the reference speed reliably;

- have small steady-state error;
- avoid excessive overshoot;
- recover from small disturbances;
- avoid long PWM saturation;
- behave similarly across repeated trials.

This is a practical, project-level definition of robustness.

7. Disturbance Testing

A disturbance is a change that affects the motor but is not directly commanded.
Examples:

- small added load;
- change in wheel contact;
- supply variation;
- friction change.

In the lab, any disturbance demonstration must be safe and instructor-approved.
Do not touch the spinning wheel directly.

8. Final Controller Evaluation

Your final controller should be evaluated using data.
Recommended metrics:

Metric	Meaning
Final speed	tracking result
Steady-state error	final accuracy
Overshoot	excessive correction
Settling time	response speed
Maximum PWM	actuator effort
PWM smoothness	controller behavior
Disturbance recovery	robustness

9. Comparing Controller Versions

By this point, your team should have tested:

- open-loop PWM;
- first-order model;
- P control;
- PI control;
- PI with filtering.

The final controller does not need to be perfect.
It should be understandable, explainable, and supported by data.

10. Preparing for the Final Presentation

A good final project explanation includes:

1. hardware setup;
2. encoder measurement;
3. open-loop behavior;
4. motor model from Week 2;
5. P-control results from Week 3;
6. PI-control results from Week 4;
7. final controller and performance;
8. what worked and what did not.

11. Check Your Understanding

Answer briefly.

1. What does $\alpha = 1$ mean in the filter?
2. Why can too much filtering be harmful?
3. Name two sources of RPM measurement noise.
4. What is one safe way to test disturbance response?
5. What data would you show to prove that your controller improved?

Summary

- Encoder speed measurements can be noisy.
- Exponential filtering uses
$$y_f[k] = \alpha y[k] + (1 - \alpha) y_f[k - 1].$$
- Smaller α gives more smoothing but more delay.
- Robustness means reliable performance under small changes.
- Final controller choice should be based on measured data.
- Week 6 will focus on final testing, refinement, and preparation for presentation.